UI / UX Design Document

Intelligent Knowledge Retrieval System

v1.0 | April 2026 | **Nexora GmbH** Internal Demo

1. Design Philosophy

The interface targets a sleek, modern SaaS aesthetic. It feels like a premium internal tool, not a prototype. Every screen is fully responsive — it works equally well on a large desktop monitor, a laptop, a tablet, and a mobile phone. Every screen should load fast, feel responsive, and communicate system state clearly.

Bright and confident

- Vivid accent colours on cards and borders
- White surfaces with coloured left-border accents
- Colour encodes role and state, not just decoration

Card-first layout

- All content lives inside rounded cards
- Clear visual grouping through padding and shadow
- Segmented panels, never raw uncontained text

Motion and feedback

- Smooth page transitions and panel slides
- Subtle entrance animations on card load
- Skeleton loaders, never blank white screens

2. Colour System

Colour carries meaning consistently across all screens.

Role	Colour	Where used
Primary	#0D9488 Teal	Nav accent, CTA buttons, active states, section heading borders
User actions	#4F46E5 Indigo	Chat send button, user message bubbles, section headings
Admin	#7C3AED Violet	Admin badge, admin-only panels and card left borders
Success	#059669 Green	Upload complete, document indexed, user created confirmations
Warning	#D97706 Amber	Partial failures, re-index needed, rate limit approaching
Error	#E11D48 Rose	Failed uploads, auth errors, query failures
Info	#0284C7 Sky	Neutral system messages, tooltips, helper text
Surface	#F3F4F6 Gray	Page background, sidebar background

Role	Colour	Where used
Card	#FFFFFF White	All card surfaces, modal backgrounds

3. Global Layout and Navigation

All authenticated screens share a fixed sidebar layout.

Sidebar (240px fixed)

- Logo and app name at top
- Nav items with icon and label
- Active item: teal left border + light teal background
- Role badge at bottom (Admin / User)
- Logout button at very bottom
- Collapses to icon-only on tablet (768px breakpoint)
- Hidden off-canvas on mobile — hamburger menu toggles it
- Overlay backdrop when open on mobile, tap outside to close

Main content area

- White card surface, 24px padding inside cards
- Page title and subtitle at top of each view
- Breadcrumb for nested screens
- Notification bell in top-right header bar
- User avatar with dropdown (profile, logout)
- Max content width 1100px, centred on large screens
- Full width with 16px horizontal padding on mobile

4. Shared Components

4.1 Cards

Every piece of content lives in a card. Cards are white with 16px border-radius and a subtle box shadow. Accent borders communicate context.

Standard card

- White surface
- 1px border all sides
- 16px radius
- 16–24px padding

Accented card

- Coloured 4px left border
- Colour matches content type
- Used for result chunks, stats
- Hover lifts slightly

Alert card

- Full-width banner style
- Coloured left border + tinted bg
- Icon on left, dismiss on right
- Stacks below top nav bar

4.2 Alerts and Toasts

Inline alerts appear inside the relevant panel. Toasts appear top-right for transient events.

> Success — Document 'Remote Work Policy.pdf' indexed successfully. 42 chunks created.

Warning — Render database expires in 12 days. Download a backup or run the seed script to restore.

Error — Upload failed. File exceeds 10MB limit or is password-protected. Try again with a valid PDF.

Info — Re-indexing in progress. Queries will use the previous index until the process completes.

4.3 Loading States

Nothing ever shows a blank white screen. Every async operation has a loading state.

Skeleton loader

- Animated grey shimmer bars
- Matches shape of incoming content
- Used on initial page load and data fetch

Progress bar upload

- Thin teal bar across top of upload card
- Shows percentage for file uploads
- Pulses while processing chunks

Spinner overlay

- Centred teal spinner on card surface
- Card dims slightly, not whole screen
- Used for query submission

4.4 Buttons

Primary

- Teal fill, white label, 8px radius
- Hover: darkens 10%, subtle lift shadow
- Active: scale down slightly
- Disabled: 40% opacity, not clickable

Secondary / Ghost

- Transparent fill, indigo border and label
- Hover: light indigo background tint
- Used alongside primary for cancel / back
- Destructive variant uses rose colour

5. Chat Interface (All Users)

The chat UI is the centrepiece. It should feel as natural as a modern messaging app while making the AI source citations visible and useful.

Message layout

- User messages: right-aligned, indigo bubble
- AI responses: left-aligned, white card with teal left border
- Timestamps shown on hover, not permanently
- Smooth slide-in animation per message
- Auto-scrolls to latest, scroll-to-top button appears

Source citations

- Each AI answer has collapsible 'Sources' section
- Source chips show document name and page
- Click chip to expand the retrieved chunk text
- Relevance score shown as a subtle percentage badge
- Multiple sources stack as horizontal chips

Input bar

- Fixed to bottom of chat panel
- Multi-line textarea, expands up to 4 rows
- Send on Enter, Shift+Enter for new line
- Character count appears near limit
- Date range filter pill above input — filters retrieval scope

Feedback and actions

- Thumbs up / down on each AI message
- Copy answer to clipboard button on hover
- Export answer + sources as .txt file download
- Clear conversation button with confirmation modal

> Chat quality signal: If the system has low confidence or no relevant chunks are found, it shows a soft amber warning inside the response card rather than hallucinating an answer. This is a visible design decision worth calling out in the README.

6. Document Ingestion (Admin)

The ingestion screen should feel effortless. Drag-and-drop first, with fallback to file picker.

Upload zone

- Large dashed-border drop zone, teal on hover
- Accepts single PDF or batch multi-select (PDF only)
- Shows file name, size, and type icon on drop
- Animated pulse while files are being read
- Reject non-PDF with immediate inline error

Upload queue

- Each file appears as a card in a vertical list
- Progress bar per file during indexing
- Status chip: Queued / Processing / Indexed / Failed
- Failed files show error reason and retry button
- Remove button to cancel before processing

Document metadata form

- Shown after file selection, before upload
- Fields: Category, Owner department, Effective date
- Optional: tags, audience (All / Engineering / etc.)
- Applied to all files in a batch or individually

Indexed documents list

- Table view of all indexed documents
- Columns: Name, Category, Owner, Date, Chunks, Status
- Search and filter by category or owner
- Delete with confirmation modal, triggers re-index
- Re-index button per document or bulk select

7. Admin Panel Screens

7.1 User Management

Screen / View	What It Contains	Key UI Notes
User list	Table of all accounts. Columns: Name, Email, Role, Last active, Status (Active / Suspended).	Search bar, role filter dropdown, sort by last active.
User detail	Slide-in side panel showing account info, query history count, and role assignment.	Edit role, suspend/reactivate account, reset password link.
Create user	Modal form: name, email, role selector. Password auto-generated and shown once.	Violet CTA button. Success toast on creation.

7.2 Query Logs

Screen / View	What It Contains	Key UI Notes
Logs table	All queries: user, timestamp, query text (truncated), chunks retrieved, feedback received.	Filter by user, date range, feedback type. Export as CSV.
Log detail	Expands a row to show full query, full AI response, all retrieved chunks with scores.	Used to audit retrieval quality and diagnose failures.

7.3 System Health

Screen / View	What It Contains	Key UI Notes
Stats cards	Row of metric cards: Total documents, Total chunks, Avg retrieval score, Queries today.	Each card has a subtle sparkline or trend arrow.
Index status	Shows last indexed timestamp, vector count, and Pinecone index health.	Re-index all button with confirmation. Status chip: Healthy / Stale / Rebuilding.

8. Authentication Screens

Login is the first impression. It should feel polished, not default.

Login page

- Centred card on a soft gradient background
- Logo and app name above the form
- Email and password inputs, clean and large
- Show/hide password toggle

Loading state

- Button shows spinner and 'Signing in...'
- Inputs disabled during submission
- No page flicker or layout shift
- Smooth fade transition to dashboard on success

Session expiry

- Amber banner on any authenticated screen
- 'Your session has expired. Please log in again.'
- Clicking CTA returns to login with redirect back
- Form state preserved where possible

- Teal primary button — full width
- Error state: rose border on inputs + alert card

9. Responsive Design

The UI is built mobile-first using CSS media queries and a fluid grid. All layouts reflow cleanly across four breakpoints.

Breakpoint	Target	Layout behaviour
≥ 1280px	**Large desktop**	Full sidebar (240px) always visible. Two and three column card grids. Max content width 1100px centred.
1024–1279px	**Laptop**	Full sidebar visible. Two column grids. Content width fluid up to container.
768–1023px	**Tablet**	Sidebar collapses to icon-only (64px). Single column grids on most views. Chat panel takes full remaining width.
< 768px	**Mobile**	Sidebar hidden off-canvas, toggle via hamburger. All grids single column. Chat input bar full width. Tables scroll horizontally inside a wrapper. Admin tables simplified to key columns only.

Chat (mobile)

- Full screen, no sidebar visible
- Input bar pinned to bottom, full width
- Source chips wrap vertically if needed
- Date filter opens as a modal sheet

Ingestion (mobile)

- Drop zone becomes a large tap target
- File picker opens native file browser
- Upload queue stacks full width
- Metadata form fields stack vertically

Admin tables (mobile)

- Priority columns shown, others hidden
- Row tap opens detail slide-up panel
- Horizontal scroll wrapper on wide tables
- Filter controls collapse into a filter button

> Touch targets: All interactive elements on mobile meet the 44px minimum touch target size. Buttons, chips, and nav icons have sufficient padding even if the visible element is smaller.

10. Animation and Motion

Motion is purposeful, not decorative. It communicates state change and guides attention.

Element	Animation	Timing
Page navigation	Fade out + slide left, new page fades in from right	200ms ease-in-out

Element	Animation	Timing
Cards on page load	Staggered fade-up, each card 40ms delayed	300ms ease-out per card
Chat message in	Slide up from bottom with fade	180ms ease-out
Skeleton loaders	Shimmer sweep left to right, looping	1.4s linear loop
File drop zone	Border pulses teal, background tints lightly	On drag-over, instant
Upload progress bar	Smooth width transition	Continuous, reflects real progress
Toast notifications	Slide in from top-right, auto-dismiss after 4s	Slide 200ms, dismiss 300ms
Modal open/close	Scale from 0.95 to 1.0 with fade	150ms ease-out
Source chip expand	Height expand with fade-in of text	200ms ease